

Appendix: The Crest Factor, the Centring Criterion, and the Diffraction Efficiency of a Multitone AOD

This appendix collects, in one place and in a single consistent notation, the four results the main text uses: what the crest factor is (Sec. 1), why it acts as a hardware constraint (Sec. 2), why the centring criterion forces the *worst possible* crest factor whenever an axis is driven with three tones (Sec. 3), and what that costs in diffracted light (Secs. 4 and 5).

On sources. Two kinds of statement appear below. The *starting points* are taken from the literature and cited: the definition and bounds of the crest factor [1, 2], the back-off of a compression-limited amplifier [3, 4], the acousto-optic diffraction law [5, 6], the coupled-wave formalism [7, 8], its multifrequency use [9], and the device data [10, 11]. The *derivations* that follow from them – the centring criterion, the $N = 3$ result, and the crest-to-efficiency relation of Sec. 5 – are carried out here in full and are self-contained; they are cited only where they rest on one of the above.

1 Definitions and conventions

1.1 The RF signal and its envelope

Each AOD axis is driven by one AWG channel carrying N tones,

$$s(t) = \sum_{n=0}^{N-1} a_n \cos(2\pi f_n t + \varphi_n), \quad f_n = f_{\text{off}} + n d, \quad (1)$$

with the tone spacing d and the tone phases φ_n , which are free parameters at the AWG. Writing $\bar{f}$ for the mean tone frequency and introducing the *centred tone index*

$$\hat{n} := n - \frac{N-1}{2} \quad \implies \quad f_n - \bar{f} = d \hat{n}, \quad (2)$$

the signal separates into a carrier at $\bar{f}$ and a complex envelope

$$\boxed{s(t) = \text{Re}\left[A(t) e^{i2\pi\bar{f}t}\right], \quad A(t) = \sum_{n=0}^{N-1} a_n e^{i(2\pi d \hat{n} t + \varphi_n)}. \quad (3)}$$

$A(t)$ carries everything the phases do. Its period is $T_{\text{env}} = 1/d$: for odd N the index $\hat{n}$ is an integer and A itself is T_{env} -periodic; for even N it is half-integer and $A(t + T_{\text{env}}) = -A(t)$, so A is *antiperiodic* while $|A|$ is still periodic with T_{env} . Only $|A|$ enters what follows, so the distinction never matters – but the averaging window below must be T_{env} in both cases.

1.2 Amplitude versus power: the one convention to get right

Two quantities in this text are easy to confuse, and they differ by a square root. The profile parameters r_x and r_y are *power* ratios – they state how much diffracted *intensity* the edge tones receive relative to the inner ones. The a_n in Eq. (1) are RF *voltage* amplitudes. Since diffracted intensity scales with RF power and power scales with the square of voltage, the edge tones carry

$$\varrho := \sqrt{r_x}, \quad a = [\varrho, 1, \dots, 1, \varrho], \quad P_n \propto a_n^2. \quad (4)$$

At the working point $r_x = 0.97$ but $\varrho = \sqrt{0.97} = 0.984886$. Every formula below is written in the a_n , i.e. in voltages.

1.3 The crest factor

$$C := \frac{\max_t |s(t)|}{\text{rms}_t[s(t)]}. \quad (5)$$

This is the standard definition for multitone signals [1, 2]; in communications engineering the square C^2 is the peak-to-average power ratio. It is a dimensionless property of the *waveform*: how far the instantaneous voltage strays above its own effective value. Both operations run over a full envelope period T_{env} .

Because the carrier at $\bar{f}$ oscillates much faster than the envelope, the two time scales separate. Locally, a sinusoid of amplitude $|A|$ contributes $\frac{1}{2}|A|^2$ to the mean square, so $\langle s^2 \rangle = \frac{1}{2}\langle |A|^2 \rangle$, while the peak of $|s|$ is the peak of $|A|$. Hence the carrier-free form

$$C = \sqrt{2} \frac{\max_t |A(t)|}{\text{rms}_t |A(t)|} = \sqrt{2} \frac{\max_t |A(t)|}{\sqrt{\sum_n a_n^2}}. \quad (6)$$

The denominator does not depend on the phases. This is what makes the crest factor tractable. Expanding,

$$\langle |A|^2 \rangle = \sum_{n,m} a_n a_m \left\langle e^{i[2\pi(f_n - f_m)t + \varphi_n - \varphi_m]} \right\rangle_T, \quad (7)$$

and the average of a phasor over a window of length T is

$$\langle e^{i[2\pi\Delta f t + \Delta\varphi]} \rangle_T = e^{i\Delta\varphi} \frac{e^{i2\pi\Delta f T} - 1}{i2\pi\Delta f T}, \quad (8)$$

which vanishes *exactly* when $\Delta f T$ is a non-zero integer – a phasor completing whole turns averages to the centre of the circle *regardless of where it starts*, which is precisely why $\Delta\varphi$ drops out. With $T = T_{\text{env}} = 1/d$ every $\Delta f = f_n - f_m$ is an integer multiple of d , so all off-diagonal terms vanish and

$$\text{rms}_t |A| = \sqrt{\sum_n a_n^2} \quad \text{for every phase set.} \quad (9)$$

The phases act on the crest factor through $\max_t |A|$ alone. If the window is chosen shorter – a common mistake – Eq. (8) no longer vanishes, the double sum survives, and the rms becomes phase-dependent.

Bounds. By the triangle inequality $|A(t)| \leq \sum_n a_n$ at all times, and $|A| \geq$ its own rms somewhere, so

$$\sqrt{2} \leq C \leq \sqrt{2} \frac{\sum_n a_n}{\sqrt{\sum_n a_n^2}} \xrightarrow{a_n \equiv 1} \sqrt{2N}. \quad (10)$$

The upper bound is attained exactly when all phasors align at some instant; the lower bound would require $|A|$ to be constant. Which phase sets approach the lower bound is a classical question: Schroeder's quadratic phases $\varphi_n = -\pi n^2/N$ reach $C \approx 1.7$ for large N [1], and the attainable minimum has been studied systematically in [2]. *None of these sets satisfies the centring criterion of Sec. 3*, which is the entire tension this appendix is about.

2 Why the crest factor is a constraint

The crest factor forbids no phase set: every φ_n is dialable at the AWG and the optical calculation is equally valid for all of them. What it limits is the *average RF power* the chain can still transmit linearly.

A power amplifier driven beyond its compression point clips the *instantaneous voltage* [4, Sec. 10.3]; it does not care about the time-averaged power until thermal limits are reached, which happens much later. With $V_{\text{peak}} = C V_{\text{rms}}$ and $P_{\text{avg}} = V_{\text{rms}}^2/R$, the peak envelope power is $P_{\text{peak}} = C^2 P_{\text{avg}}$, and demanding $P_{\text{peak}} \leq P_{\text{1dB}}$ gives

$$\boxed{P_{\text{avg}} \leq \frac{P_{\text{1dB}}}{C^2}}. \quad (11)$$

The 1 dB compression point, not the saturated power, is the right threshold: linearity is exactly what the field calculation assumes. Backing a multitone transmitter off by its peak-to-average ratio is standard practice in power-amplifier design [3]; Eq. (11) is that rule written for this chain.

Three budgets must be kept apart:

Limit	belongs to	depends on C ?
2 W average RF power	AOD transducer (heating)	no
Compression	amplifier (peak voltage)	yes , $\propto 1/C^2$
Diffraction saturation $\sin^2$	AOD (total RF power)	no

With the amplifier used here [11], $P_{\text{1dB}} = 3.98 \text{ W}$, the crest factors of the working point give $P_{\text{avg}} \leq 0.66 \text{ W}$ on the x axis and 0.83 W on y – well below the AOD's 2 W. *The amplifier is the limit, not the AOD*, and Eq. (11) says the crest factor is the exchange rate.

3 The centring criterion

3.1 What "centred" has to mean

The atom sits at one place, smeared by its thermal motion over $\sigma \approx 108 \text{ nm}$; the thermal position distribution in a tweezer and its measurement are described in [12]. What harms it is not that the interference pattern beats, and not that the pulse is dim: it is that the bright spot can sit *off to one side* during the pulse. The pulse area is then larger on one edge of the position distribution than on the other, the rotation angle depends on where the atom happened to be, and no calibration of the mean can repair it.

The requirement is therefore not "the pattern is symmetric at some instant" but

$$I(-\mathbf{r}, t) = I(\mathbf{r}, t) \quad \text{for all } t, \quad (12)$$

with the atom at $\mathbf{r} = 0$. This implies $\nabla I|_0 = 0$ at all times: the intensity has a stationary point at the atom throughout the pulse.

3.2 The three symmetries of the profile

The field in the focal plane is a sum over the $N_x \times N_y$ spots,

$$E(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{-i2\pi(f_n + g_m)t} e^{i(\varphi_n + \psi_m)}, \quad (13)$$

with u the (even) single-spot envelope, $\mathbf{c}_{nm}$ the spot positions, and g_m, ψ_m the y -axis frequencies and phases. Writing $\bar{n} := N - 1 - n$ for the mirror index, the profile is built symmetric in three independent ways:

$$\mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{c}_{nm} \quad (\text{positions: the spot grid is centred}), \quad (14)$$

$$a_{\bar{n}} = a_n, \quad b_{\bar{m}} = b_m \quad (\text{amplitudes: } [\varrho, 1, \dots, 1, \varrho]), \quad (15)$$

$$f_{\bar{n}} = F - f_n, \quad g_{\bar{m}} = G - g_m \quad (\text{frequencies}). \quad (16)$$

Equation (16) follows directly from the equidistant raster: $f_n = f_{\text{off}} + n d$ gives $f_n + f_{\bar{n}} = 2f_{\text{off}} + (N - 1)d =: F$, independent of n .

3.3 Derivation

Apply the three symmetries to $E(-\mathbf{r}, t)$. Relabel the sum by the mirror index, $n \rightarrow \bar{n}$, $m \rightarrow \bar{m}$ – a permutation, so nothing is lost:

$$E(-\mathbf{r}, t) = \sum_{n,m} a_{\bar{n}} b_{\bar{m}} u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}}) e^{-i2\pi(f_{\bar{n}} + g_{\bar{m}})t} e^{i(\varphi_{\bar{n}} + \psi_{\bar{m}})}. \quad (17)$$

Now insert the three symmetries one at a time.

Step 1 (positions, and u even). By Eq. (14), $-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}} = -\mathbf{r} + \mathbf{c}_{nm} = -(\mathbf{r} - \mathbf{c}_{nm})$, and since u is even, $u(-\mathbf{r} - \mathbf{c}_{\bar{n}\bar{m}}) = u(\mathbf{r} - \mathbf{c}_{nm})$ – the same envelope factor as in $E(\mathbf{r}, t)$.

Step 2 (amplitudes). By Eq. (15), $a_{\bar{n}} b_{\bar{m}} = a_n b_m$: the amplitudes are unchanged.

Step 3 (frequencies). This is the decisive one. By Eq. (16),

$$e^{-i2\pi(f_{\bar{n}} + g_{\bar{m}})t} = e^{-i2\pi(F+G)t} \underbrace{e^{+i2\pi(f_n + g_m)t}}_{\text{the conjugate of the original factor}}. \quad (18)$$

The frequency mirror does not reproduce the time factor – it *conjugates* it, up to one global factor $e^{-i2\pi(F+G)t}$ that is the same for every spot.

Step 4 (collect). Putting the three together,

$$E(-\mathbf{r}, t) = e^{-i2\pi(F+G)t} \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n + g_m)t} e^{i(\varphi_n + \psi_m)}, \quad (19)$$

which is to be compared with

$$E^*(\mathbf{r}, t) = \sum_{n,m} a_n b_m u(\mathbf{r} - \mathbf{c}_{nm}) e^{+i2\pi(f_n + g_m)t} e^{-i(\varphi_n + \psi_m)}. \quad (20)$$

The two sums run over the same basis functions with the same real coefficients, and differ only in a constant phase factor per term. They are therefore proportional – $E(-\mathbf{r}, t) = e^{i\alpha(t)} E^*(\mathbf{r}, t)$, hence $I(-\mathbf{r}, t) = I(\mathbf{r}, t)$ – if and only if

$$e^{i(\varphi_{\bar{n}} + \psi_{\bar{m}})} = e^{i\Theta} e^{-i(\varphi_n + \psi_m)} \quad \text{for all } n, m \quad (21)$$

with a *single* constant Θ independent of n and m . Splitting the n - and m -dependence gives the criterion:

$$\boxed{\varphi_n + \varphi_{N-1-n} = c \quad \text{for all } n, \quad \psi_m + \psi_{M-1-m} = c' \quad \text{for all } m.} \quad (22)$$

3.4 Conjugate, not mirrored

Equation (22) is often misread as "the phases must be symmetric", $\varphi_{\bar{n}} = \varphi_n$. That is a *different* condition and it does not centre the pattern. The mirror in Eq. (16) conjugates the time factor, so the phases must *anti*-compensate, i.e. sum to a constant. Numerically, mirror-symmetric phase sets leave centroid offsets of 44 nm to 84 nm against $\sigma = 108$ nm, while conjugate-symmetric sets give 0.00 nm. The two coincide only for purely real phasors, $\varphi_n \in \{0, \pi\}$, which is the source of the confusion.

3.5 The middle tone is its own mirror partner

A natural objection: *for odd N the middle tone has no partner, so its phase is unconstrained*. It is not. The mirror acts on the indices, $n \mapsto \bar{n} = N - 1 - n$; this is an involution, and involutions have pairs *and fixed points*:

N	pairing	fixed point
3	$0 \leftrightarrow 2, \quad 1 \mapsto 1$	$n = 1$
4	$0 \leftrightarrow 3, \quad 1 \leftrightarrow 2$	none
5	$0 \leftrightarrow 4, \quad 1 \leftrightarrow 3, \quad 2 \mapsto 2$	$n = 2$

A fixed point is not an unpaired element: it is an element paired *with itself*. Equation (22) is required for every n , so at the fixed point it reads $\varphi_n + \varphi_n = c$ and

$$\boxed{\varphi_{(N-1)/2} = \frac{c}{2} \quad \text{or} \quad \frac{c}{2} + \pi \quad (N \text{ odd}).} \quad (23)$$

The two roots appear because Eq. (22) holds modulo 2π .

The objection implicitly assumes each pair may carry its own constant. It may not: c is the single global Θ of the derivation, and a single $e^{i\alpha(t)}$ has to serve all spots at once, or the contributions do not add up to a conjugated field. The middle tone sits at the centre frequency, so its spot lies at the mirror centre; under the mirror it maps onto itself, and its condition is not "none" but "with itself".

Numerical check. With $\varphi_x = (0, \psi, 0)$ – so $c = 0$ – and the y phases held fixed at $(0; 98; 16; 114)^\circ$, which satisfy their own criterion ($0 + 114 = 98 + 16$), the largest intensity gradient at the atom over a full beat period is

ψ	$\max_t \partial_x I /I$	$\max_t \partial_y I /I$	
0°	$4 \cdot 10^{-10}$	$4 \cdot 10^{-10}$	criterion satisfied
10°	$8.8 \cdot 10^{-4}$	$2.6 \cdot 10^{-4}$	
30°	$2.5 \cdot 10^{-3}$	$7.4 \cdot 10^{-4}$	
90°	$5.1 \cdot 10^{-3}$	$1.5 \cdot 10^{-3}$	
180°	$4 \cdot 10^{-10}$	$4 \cdot 10^{-10}$	criterion satisfied

The gradient vanishes (to the noise floor of the numerical derivative) at exactly the two values predicted by Eq. (23) and nowhere else; with $c = 60^\circ$ the zeros move to $\psi = 30^\circ$ and 210° . Note the third column: violating the criterion in x also tilts the y gradient. With a separable (Gaussian) envelope it would not; the real Airy envelope is radial and couples the axes, so the criterion must hold on *both*.

3.6 Structure of the admissible family

Writing $\varphi_n = c/2 + \chi_n$, Eq. (22) becomes

$$\chi_{\bar{n}} = -\chi_n, \quad (24)$$

i.e. a global phase plus an *antisymmetric* part. Antisymmetric vectors in $\mathbb{R}^N$ form a subspace of dimension $\lfloor N/2 \rfloor$, so

$$\dim(\text{family}) = \left\lfloor \frac{N}{2} \right\rfloor \quad \text{instead of} \quad N - 1. \quad (25)$$

The criterion halves the phase freedom. One of the remaining dimensions is moreover crest-neutral: a *ramp* $\chi_n = \delta \hat{n}$ is antisymmetric (since $\hat{\bar{n}} = -\hat{n}$), so every ramp satisfies the criterion automatically, and substituting it into Eq. (3) gives

$$A_\delta(t) = A_{\delta=0}(t + \tau), \quad \tau = \frac{\delta}{2\pi d}, \quad (26)$$

a pure time shift. Maximum and rms over a full period are invariant under a shift, so a ramp cannot change C . Hence

$$\dim(\text{crest-relevant}) = \left\lfloor \frac{N}{2} \right\rfloor - 1. \quad (27)$$

For $N = 3$ this is **zero**.

3.7 Why $N = 3$ always gives the worst crest factor

The counting already says there is nothing left to vary, but the direct calculation is shorter and needs no ramp argument at all. With $\theta := 2\pi d t$ and the voltage amplitudes of Eq. (4), $a_0 = a_2 = \varrho$, $a_1 = 1$:

$$A(t) = \varrho e^{i(\varphi_0 - \theta)} + e^{i\varphi_1} + \varrho e^{i(\varphi_2 + \theta)}. \quad (28)$$

The criterion requires $\varphi_0 + \varphi_2 = c$ and $2\varphi_1 = c$, so the two outer phases lie symmetrically about $c/2$:

$$\varphi_0 = \frac{c}{2} - \chi, \quad \varphi_2 = \frac{c}{2} + \chi, \quad \varphi_1 = \frac{c}{2} \text{ or } \frac{c}{2} + \pi, \quad (29)$$

with χ the only remaining parameter. Factoring out $e^{ic/2}$,

$$\boxed{A(t) e^{-ic/2} = \varrho e^{-i(\chi + \theta)} \pm 1 + \varrho e^{+i(\chi + \theta)} = 2\varrho \cos(\theta + \chi) \pm 1.} \quad (30)$$

The sign distinguishes the two branches for φ_1 . The entire time dependence now sits in a *real* cosine, which sweeps the full interval $[-1, 1]$ over one period no matter what χ is – χ only sets *when*. Maximising over $u := \cos(\theta + \chi) \in [-1, 1]$,

$$\text{branch } + : \quad \max_u |2\varrho u + 1| = 2\varrho + 1 \quad \text{at } u = +1, \quad (31)$$

$$\text{branch } - : \quad \max_u |2\varrho u - 1| = |-2\varrho - 1| = 2\varrho + 1 \quad \text{at } u = -1, \quad (32)$$

so with $a_0 + a_1 + a_2 = 2\varrho + 1$,

$$\max_t |A| = \sum_n a_n \quad \text{for both branches and every } \chi, c. \quad (33)$$

But $\sum_n a_n$ is the triangle-inequality bound of Eq. (10), valid for *any* phase set whatsoever. The admissible family therefore always attains it:

With three tones, the centring criterion fixes the crest factor at the largest value the amplitudes allow, for every c , every χ and both branches. This axis cannot be relieved by any choice of phase. The reason is visible in Eq. (30): the only free parameter appears *inside* the cosine. It can move *when* the phasors align, not *whether* they do.

Numerically, with $\varrho = \sqrt{0.97} = 0.9848858$:

$$\sum_n a_n = 2\varrho + 1 = 2.9697716, \quad \sqrt{\sum_n a_n^2} = \sqrt{2r_x + 1} = \sqrt{2.94} = 1.7146428, \quad (34)$$

– note that the square sum contains the *power* ratio linearly, the square root of Eq. (4) cancelling
– and therefore

$$C_x = \sqrt{2} \frac{2.9697716}{1.7146428} = 2.4494263. \quad (35)$$

For exactly equal amplitudes this would be $\sqrt{2N} = \sqrt{6} = 2.4494897$; the working point lies $6 \cdot 10^{-5}$ below it, so the two agree to four digits without being the same number.

3.8 Contrast: $N = 4$

Here Eq. (22) leaves $\varphi_y = [0, b, c - b, c]$ with *two* parameters. The ramp is the special case $b = c/3$; any other b is an antisymmetric deviation that is not a ramp and therefore genuinely reshapes the envelope. The two inner tones can then no longer be aligned with the outer ones simultaneously, $\max |A|$ drops below $\sum_n a_n$, and with it the crest factor. At $c = 114^\circ$:

b	φ_y [deg]	$\max A $	C_y	
38°	$[0, 38, 76, 114]$	4.154	2.8265	(ramp, $= \sum a_n$)
60°	$[0, 60, 54, 114]$	4.022	2.7364	
98°	$[0, 98, 16, 114]$	3.228	2.1963	
114°	$[0, 114, 0, 114]$	3.639	2.4761	

That single parameter is worth 65 % more average RF power through Eq. (11) – and because the x axis is fixed, it is the *only* adjustment the criterion leaves. The structural reason is Eq. (27) together with the fixed-point argument: even N has no self-paired tone, so no individual phase is pinned.

4 From RF power to diffracted light

4.1 One grating: where the $\sin^2$ and the $\sqrt{P}$ come from

The RF drives a piezo transducer; the acoustic strain amplitude S obeys $P_{ac} \propto S^2$, and the photoelastic effect turns strain into an index modulation [5, Ch. 20]

$$\Delta n = -\frac{1}{2} n_0^3 p S \propto \sqrt{P_{ac}}. \quad (36)$$

This is the origin of the square root – it lies in the conversion from power to amplitude, not in the diffraction. In the Bragg regime the incident and diffracted waves exchange amplitude periodically along the interaction length, giving $\eta = \sin^2 \nu$ with $\nu = \pi \Delta n L / (\lambda \cos \theta_B)$, hence

$$\boxed{\eta(P) = \sin^2(\kappa \sqrt{P}), \quad \kappa = \frac{\pi}{\lambda} \sqrt{\frac{M_2 L}{2H}} \sqrt{\zeta},} \quad (37)$$

with the figure of merit M_2 , the transducer aspect ratio L/H and the electro-acoustic conversion efficiency ζ . Equation (37) is the standard Bragg-regime result [5, Ch. 20] and is quoted in exactly

this form in the manufacturer's application notes [6]. None of the four is known individually to useful accuracy, which is why κ is *measured*: one efficiency point at a known power replaces all of them. From the test sheet [10] (58 % in order "1-1" at 1.3 W per axis, so $\sqrt{0.58} = 0.762$ per axis),

$$\kappa = \frac{\arcsin \sqrt{0.762}}{\sqrt{1.3}} = 0.930 \sqrt{W}. \quad (38)$$

4.2 Many tones: the coupled-mode equations

Write the field inside the crystal as one undiffracted wave plus N diffracted ones, each with a slowly varying amplitude:

$$E = A_u(z) e^{i(\mathbf{k}_u \mathbf{r} - \omega_0 t)} + \sum_{n=0}^{N-1} A_n(z) e^{i(\mathbf{k}_n \mathbf{r} - \omega_n t)}, \quad \omega_n = \omega_0 + \Omega_n. \quad (39)$$

Notation: A_u carries no number – it belongs to no tone; A_n is the beam diffracted by tone n , and every $\sum_n$ below runs over the tones and never includes A_u . This is the coupled-wave ansatz of [7], extended from one grating to N ; a general treatment of the acousto-optic coupled-wave equations is given in [8]. Substituting into the Helmholtz equation with $n^2 \approx n_0^2 + 2n_0 \sum_n \Delta n_n \cos(\mathbf{K}_n \mathbf{r} - \Omega_n t + \varphi_n)$, keeping terms that oscillate at ω_n and referring them to the carrier $e^{i\mathbf{k}_n \mathbf{r}}$:

$$\frac{dA_n}{dz} = -i \kappa_n e^{i\varphi_n} A_u e^{i\Delta k_n z}, \quad \frac{dA_u}{dz} = -i \sum_n \kappa_n e^{-i\varphi_n} A_n e^{-i\Delta k_n z}, \quad (40)$$

with $\kappa_n = \kappa \sqrt{P_n}$ and the momentum mismatch $\Delta k_n = (\mathbf{k}_u + \mathbf{K}_n - \mathbf{k}_n) \cdot \hat{\mathbf{z}}$. Three things follow immediately.

(a) The tone phases drop out. The substitution $\tilde{A}_n = A_n e^{-i\varphi_n}$ removes φ_n from both equations, and $|A_n| = |\tilde{A}_n|$. **The diffraction efficiency is exactly independent of the tone phases;** φ_n reappears only as the phase of the diffracted wave, i.e. in the interference pattern, where it belongs. This is why the crest factor – a property of the *sum s(t)*, which only the broadband amplifier ever sees – cannot appear anywhere in this section.

(b) Phase matching is assumed here, not later. Setting $\Delta k_n = 0$ and transforming the phases away leaves

$$\frac{dA_u}{dz} = -i \sum_n \kappa_n A_n, \quad \frac{dA_n}{dz} = -i \kappa_n A_u, \quad (41)$$

where z is measured in units of the interaction length, so the crystal exit is at $z = 1$ and κ_n is the coupling accumulated over the whole crystal. For $\Delta k \neq 0$ the two-wave solution acquires the detuned form $\eta(\xi) = \nu^2 (\nu^2 + \xi^2)^{-1} \sin^2 \sqrt{\nu^2 + \xi^2}$ with $\nu = \kappa \sqrt{P}$ and $\xi = \Delta k L/2$ [7]; calibrating ξ against the specified 36 MHz bandwidth [10] (half efficiency at the band edge) makes the 0.37 MHz tone span of the working point worth 0.007 % in efficiency, while an 18 MHz span would already cost 15 %.

(c) Unequal Δk_n would break the reduction below, because carrying $e^{-i\Delta k_n z}$ through the derivative produces a new combination with weights $\kappa_n \Delta k_n$, and the hierarchy does not close unless all Δk_n are equal (or zero).

4.3 Reduction to two waves

Bright mode. Define

$$\kappa_{\text{eff}} = \sqrt{\sum_n \kappa_n^2}, \quad B = \frac{1}{\kappa_{\text{eff}}} \sum_n \kappa_n A_n. \quad (42)$$

Then $dA_u/dz = -i\kappa_{\text{eff}} B$ and $dB/dz = -i\kappa_{\text{eff}} A_u$: a closed two-wave system.

Dark modes stay dark. Any $C = \sum_n c_n A_n$ with $\sum_n c_n \kappa_n = 0$ obeys $dC/dz = 0$, and $C(0) = 0$ since nothing is diffracted at the entrance. So $C \equiv 0$: of the N diffracted degrees of freedom, $N - 1$ are never excited. This is what makes the reduction exact rather than approximate.

Solve and return. With $A_u(0) = 1, B(0) = 0$: $A_u = \cos(\kappa_{\text{eff}} z), B = -i \sin(\kappa_{\text{eff}} z)$. Integrating the second of Eq. (41) with $A_n(0) = 0$, the same integral appears for every n :

$$A_n(z) = -i \kappa_n \int_0^z A_u(z') dz' = -i \frac{\kappa_n}{\kappa_{\text{eff}}} \sin(\kappa_{\text{eff}} z), \quad (43)$$

so the orders differ only by the prefactor κ_n . Summing at $z = 1$,

$$\eta_{\text{tot}} = \sum_n |A_n|^2 = \sin^2 \kappa_{\text{eff}} = 1 - |A_u|^2, \quad (44)$$

and – the step where the sum moves under the root –

$$\kappa_{\text{eff}} = \sqrt{\sum_n \kappa_n^2} = \sqrt{\sum_n \kappa^2 P_n} = \kappa \sqrt{P_{\text{tot}}}. \quad (45)$$

The squares of the couplings *are* the powers, so their quadratic sum is nothing but the ordinary sum of powers under one square root. Hence

$$\boxed{\eta_{\text{tot}} = \sin^2(\kappa \sqrt{P_{\text{tot}}}), \quad \eta_n = \frac{P_n}{P_{\text{tot}}} \eta_{\text{tot}}, \quad P_{\text{tot}} = \sum_n P_n.} \quad (46)$$

Energy is conserved exactly, $|A_u|^2 + \sum_n |A_n|^2 = 1$, and for $N = 1$ Eq. (46) reduces to Eq. (37).

Relation to the literature. Multifrequency Bragg diffraction with optimised tone phases has been treated experimentally and theoretically by Antonov and Rezvov [9], who split one beam into seven equal channels at 80 % net efficiency; the same device class underlies the multitone AOD tweezer arrays of [13]. The point relevant here is the following.

The sum belongs under the root. The N gratings act as *one* grating of strength $\sqrt{\sum_n \kappa_n^2}$: an additional tone does not open a new channel out of the undiffracted beam, it shares the existing one. The naive alternative $\sum_n \sin^2(\kappa \sqrt{P_n})$ is not wrong physics but the small-signal limit taken in the wrong place – for $\kappa \sqrt{P} \ll 1$ both expressions tend to $\kappa^2 P_{\text{tot}}$. They separate exactly where we work:

P_{tot}	$\kappa \sqrt{P_{\text{tot}}}$	η_{tot} exact	naive sum (3 tones)
5 mW	0.066	0.00432	0.00433 (+0.10 %)
50 mW	0.208	0.0427	0.0431 (+0.97 %)
0.664 W	0.758	0.473	0.539 (+14.0 %)

A useful consequence: *the total diffraction efficiency depends only on the total RF power, not on how it is divided among tones.* Splitting into a multitone profile costs no efficiency at all.

5 What the crest factor costs in diffracted light

The two halves now join. Equation (46) says the diffraction depends only on P_{tot} ; Eq. (11) says the crest factor caps P_{tot} . Substituting one into the other gives the whole optical effect of the crest factor in a single expression:

$$\boxed{\eta_{\text{axis}}(C) = \sin^2\left(\frac{\kappa \sqrt{P_{\text{1dB}}}}{C}\right), \quad \eta_{\text{profile}} = \eta_x \eta_y.} \quad (47)$$

With $\kappa = 0.930\sqrt{W}$ from Eq. (38) [10] and $P_{\text{dB}} = 3.98\text{ W}$ [11] the numerator is $\kappa\sqrt{P_{\text{dB}}} = 1.8559$, so $\eta_{\text{axis}} = \sin^2(1.8559/C)$:

Table 1: The crest factor translated into diffraction, via Eq. (47).

	C	$P_{\text{avg}} = P_{\text{dB}}/C^2$	η_{axis}	
theoretical minimum	$\sqrt{2} = 1.414$	1.99 W	0.935	unreachable
x with $\psi = 90^\circ$	1.822	1.20 W	0.725	centring abandoned
y , $b = 98^\circ$	2.196	0.83 W	0.559	working point
x, centred, N = 3	2.449	0.66 W	0.472	forced, Sec. 3.7
y , ramp	2.827	0.50 W	0.373	worst case

At the working point, $\eta_{\text{profile}} = 0.472 \times 0.559 = 26.4\%$, i.e. between 2.0% and 2.4% per spot over the twelve spots. Had both axes reached the theoretical minimum $C = \sqrt{2}$, the profile efficiency would be 87% – a factor 3.3 more light. That factor is the price of the crest factor, and Sec. 3.7 says how much of it is negotiable: on the x axis, none of it, as long as three tones are used and the pattern must stay centred on the atom.

Three closing remarks.

1. **The chain is one-directional.** Phases $\rightarrow C \rightarrow P_{\text{avg}} \rightarrow \eta \rightarrow$ Rabi frequency. Nowhere do the phases act on η directly; see Sec. 4(a). The crest factor is a budget limit of the electronics that happens to propagate all the way to the atom.
2. **The remedy is more tones, not better phases.** By Eq. (27) the crest-relevant freedom is $\lfloor N/2 \rfloor - 1$, zero only for $N \leq 3$. A fourth x tone dissolves the conflict instead of negotiating it.
3. **One channel is not covered here.** Equation (37) assumes a linear transducer. The instantaneous strain in the crystal carries the crest factor C as well, so a high C also brings on acoustic nonlinearity earlier – an effect independent of the amplifier, pointing the same way, and not quantified here.

References

- [1] M. R. Schroeder. Synthesis of low-peak-factor signals and binary sequences with low autocorrelation. *IEEE Transactions on Information Theory*, 16(1):85–89, 1970. doi: 10.1109/TIT.1970.1054411.
- [2] Stephen Boyd. Multitone signals with low crest factor. *IEEE Transactions on Circuits and Systems*, 33(10):1018–1022, 1986. doi: 10.1109/TCS.1986.1085837.
- [3] Steve C. Cripps. *RF Power Amplifiers for Wireless Communications*. Artech House, Norwood, MA, 2 edition, 2006. ISBN 978-1-59693-018-6.
- [4] David M. Pozar. *Microwave Engineering*. Wiley, Hoboken, NJ, 4 edition, 2012. ISBN 978-0-470-63155-3. Ch. 10 (Noise and Nonlinear Distortion), Sec. 10.3.
- [5] Bahaa E. A. Saleh and Malvin Carl Teich. *Fundamentals of Photonics*. Wiley, Hoboken, NJ, 3 edition, 2019. ISBN 978-1-119-50687-4. Ch. 20, Acousto-Optics.
- [6] AA Opto-Electronic. Acousto-optics — application notes. <https://aaoptoelectronic.com/wp-content/uploads/2024/04/AA-THEORY-NOTES.pdf>, . accessed 17 September 2026.

- [7] H. Kogelnik. Coupled wave theory for thick hologram gratings. *Bell System Technical Journal*, 48(9):2909–2947, 1969. doi: 10.1002/j.1538-7305.1969.tb01198.x.
- [8] Vincent Laude. General solution of the coupled-wave equations of acousto-optics. *Journal of the Optical Society of America A*, 20(12):2307, 2003. doi: 10.1364/JOSAA.20.002307.
- [9] S. N. Antonov and Yu. G. Rezvov. Efficient multiple-beam Bragg acoustooptic diffraction with phase optimization of a multifrequency acoustic wave. *Technical Physics*, 52(8):1053–1060, 2007. doi: 10.1134/S1063784207080154.
- [10] AA Opto-Electronic. DTSXY-400-800 two-axis acousto-optic deflector: data sheet and individual test sheet s/n 3001-o211386, . manufacturer documentation.
- [11] Mini-Circuits. ZHL-03-5WF+ coaxial high power amplifier, data sheet. <https://www.minicircuits.com/pdfs/ZHL-03-5WF+.pdf>. accessed 17 September 2026.
- [12] C. Tuchendler, A. M. Lance, A. Browaeys, Y. R. P. Sortais, and P. Grangier. Energy distribution and cooling of a single atom in an optical tweezer. *Physical Review A*, 78(3):033425, 2008. doi: 10.1103/PhysRevA.78.033425.
- [13] M. Endres, H. Bernien, A. Keesling, H. Levine, E. R. Anschuetz, A. Krajenbrink, C. Senko, V. Vuletic, M. Greiner, and M. D. Lukin. Atom-by-atom assembly of defect-free one-dimensional cold atom arrays. *Science*, 354(6315):1024–1027, 2016. doi: 10.1126/science.aah3752.

BibTeX source

Copy the following into your own .bib file. Every entry was checked against Crossref or the publisher's page; the two data sheets are manufacturer documents.

```
@article{schroeder1970,
  author = {Schroeder, M. R.},
  title = {Synthesis of low-peak-factor signals and binary sequences
    with low autocorrelation},
  journal = {IEEE Transactions on Information Theory},
  volume = {16}, number = {1}, pages = {85--89}, year = {1970},
  doi = {10.1109/TIT.1970.1054411}
}

@article{boyd1986,
  author = {Boyd, Stephen},
  title = {Multitone signals with low crest factor},
  journal = {IEEE Transactions on Circuits and Systems},
  volume = {33}, number = {10}, pages = {1018--1022}, year = {1986},
  doi = {10.1109/TCS.1986.1085837}
}

@book{cripps2006,
  author = {Cripps, Steve C.},
  title = {{RF} Power Amplifiers for Wireless Communications},
  edition = {2}, publisher = {Artech House}, address = {Norwood, MA},
  year = {2006}, isbn = {978-1-59693-018-6}
}

@book{pozar2012,
  author = {Pozar, David M.},
  title = {Microwave Engineering},
  edition = {4}, publisher = {Wiley}, address = {Hoboken, NJ},
  year = {2012}, isbn = {978-0-470-63155-3},
  note = {Ch.~10 (Noise and Nonlinear Distortion), Sec.~10.3}
}

@book{salehteich2019,
  author = {Saleh, Bahaa E. A. and Teich, Malvin Carl},
  title = {Fundamentals of Photonics},
  edition = {3}, publisher = {Wiley}, address = {Hoboken, NJ},
  year = {2019}, isbn = {978-1-119-50687-4},
  note = {Ch.~20, Acousto-Optics}
}

@article{kogelnik1969,
  author = {Kogelnik, H.},
  title = {Coupled Wave Theory for Thick Hologram Gratings},
  journal = {Bell System Technical Journal},
  volume = {48}, number = {9}, pages = {2909--2947}, year = {1969},
  doi = {10.1002/j.1538-7305.1969.tb01198.x}
}

@article{laude2003,
  author = {Laude, Vincent},
  title = {General solution of the coupled-wave equations of acousto-optics},
  journal = {Journal of the Optical Society of America A},
  volume = {20}, number = {12}, pages = {2307}, year = {2003},
  doi = {10.1364/JOSAA.20.002307}
}

@article{antonov2007,
  author = {Antonov, S. N. and Rezvov, Yu. G.},
  title = {Efficient multiple-beam {B}ragg acoustooptic diffraction with
    phase optimization of a multifrequency acoustic wave},
  journal = {Technical Physics},
  volume = {52}, number = {8}, pages = {1053--1060}, year = {2007},
  doi = {10.1134/S1063784207080154}
}

@article{endres2016,
  author = {Endres, M. and Bernien, H. and Keesling, A. and Levine, H. and
    Anschuetz, E. R. and Krajenbrink, A. and Senko, C. and
    Vuletic, V. and Greiner, M. and Lukin, M. D.},
```

```

    title   = {Atom-by-atom assembly of defect-free one-dimensional cold atom arrays},
    journal = {Science},
    volume  = {354}, number = {6315}, pages = {1024--1027}, year = {2016},
    doi     = {10.1126/science.aah3752}
}

@article{tuchendler2008,
  author = {Tuchendler, C. and Lance, A. M. and Browaeys, A. and
    Sortais, Y. R. P. and Grangier, P.},
  title   = {Energy distribution and cooling of a single atom in an optical tweezer},
  journal = {Physical Review A},
  volume  = {78}, number = {3}, pages = {033425}, year = {2008},
  doi     = {10.1103/PhysRevA.78.033425}
}

@misc{aaoptonotes,
  author = {{AA Opto-Electronic}},
  title   = {Acousto-Optics --- Application Notes},
  howpublished = {\url{https://aaoptoelectronic.com/wp-content/uploads/2024/04/AA-THEORY-NOTES.pdf}},
  note    = {accessed 17 September 2026}
}

@misc{aaaod,
  author = {{AA Opto-Electronic}},
  title   = {{DTSXY-400-800} two-axis acousto-optic deflector:
    data sheet and individual test sheet S/N 3001-0211386},
  note    = {manufacturer documentation}
}

@misc{minicircuits,
  author = {{Mini-Circuits}},
  title   = {{ZHL-03-5WF+} coaxial high power amplifier, data sheet},
  howpublished = {\url{https://www.minicircuits.com/pdfs/ZHL-03-5WF+.pdf}},
  note    = {accessed 17 September 2026}
}

```